Exercise discussion Inertial Navigation

Ex 03
Strapdown Algorithm

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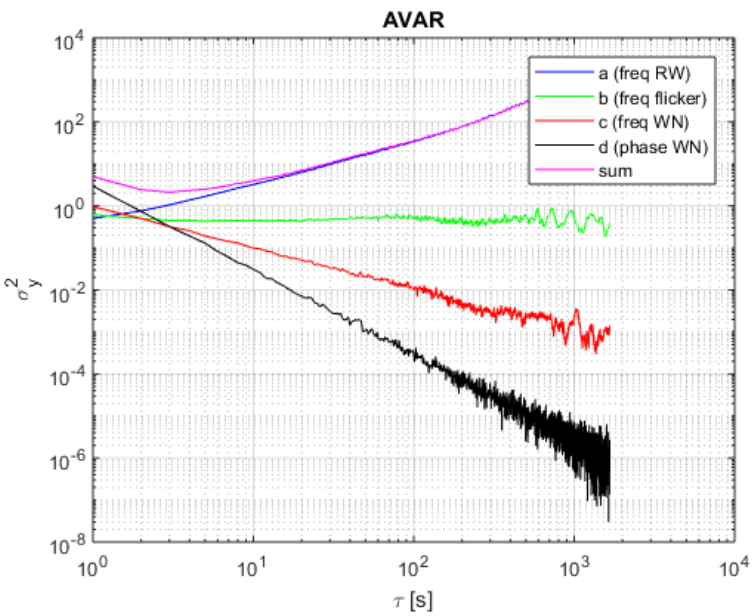
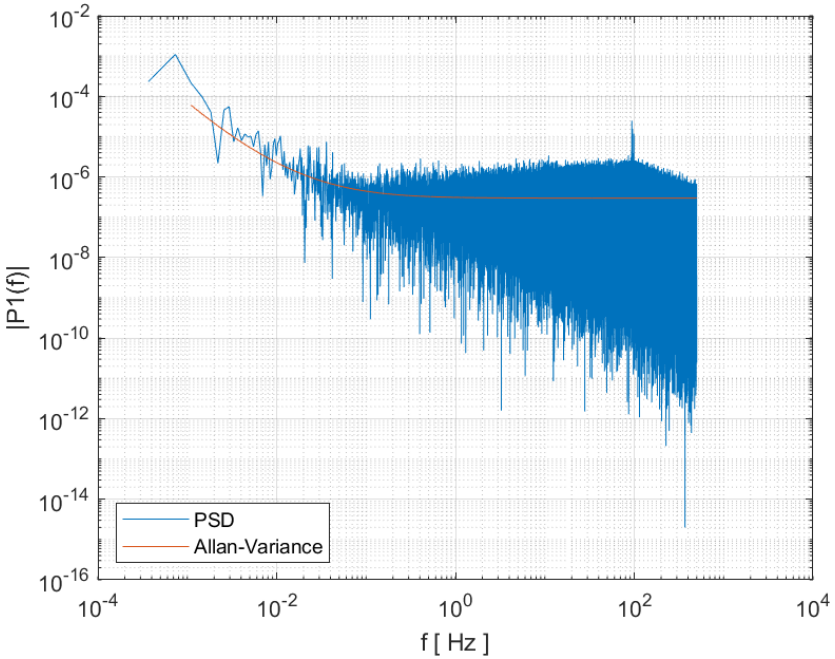
Review Exercise 02

- Task 1: ideal bias: 0, ideal scale factor: 1.
- Gyro was not sensitive enough for Earth rate → wrong scale factor estimate, bias too large





- Task 3: Acc shows random walk, gyro not (large white noise)
- Task 4: FFT yields information about frequency and amplitude of sinusoidal perturbations. Works better for high frequency signals. AVAR is superior for long periodic and deterministic trends.

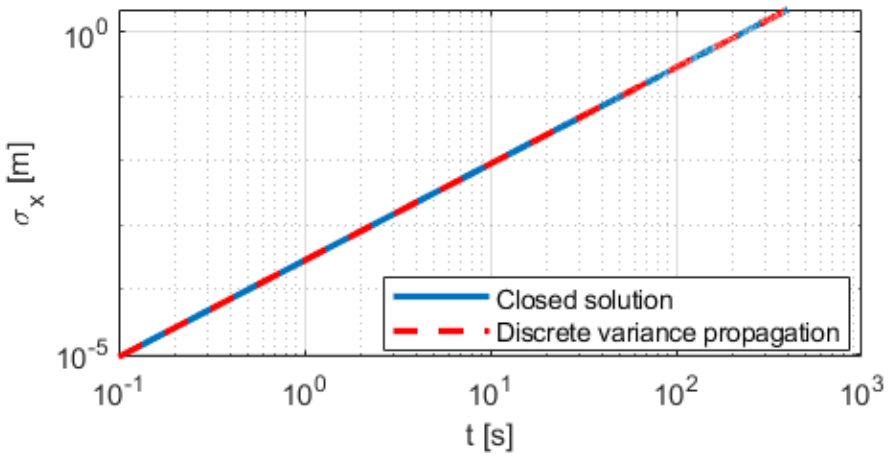
Review Exercise 02 - Task 4 continued

- LTI system, closed analytical solution possible ($t_0 = 0$):

$$\Phi(t) = \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}, \quad Q(t) = \begin{bmatrix} \frac{1}{3} h_0 t^3 & \frac{1}{2} h_0 t^2 \\ \frac{1}{2} h_0 t^2 & h_0 t \end{bmatrix}. \tag{1}$$

- Position standard deviation:

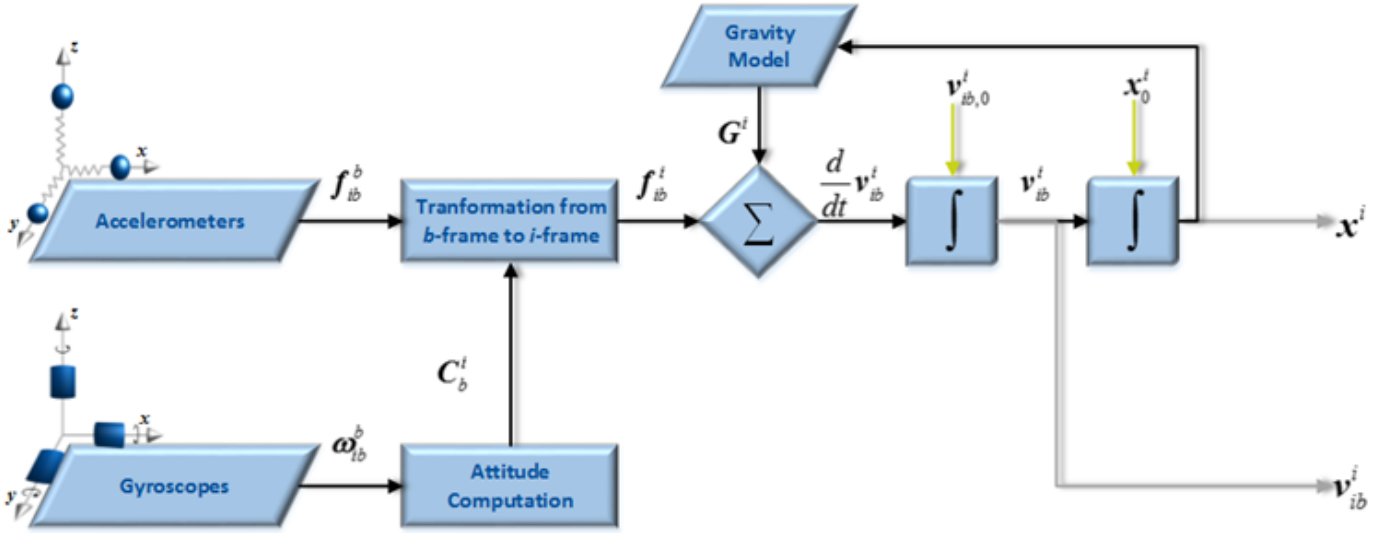
$$\sigma_x(t) = \sqrt{Q_{(1,1)}(t)}. \tag{2}$$

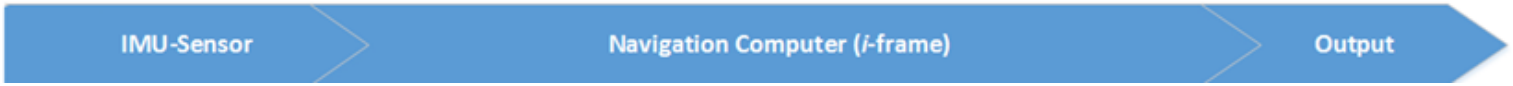


- Discussion:
  - In a **loosely coupled GNSS/INS** system, white noise density can be used to approximate the accumulated position or velocity error *between* GNSS measurements.
  - **Free inertial navigation**: undetected bias or misalignment (projection of g) often have larger impact → calibration important.

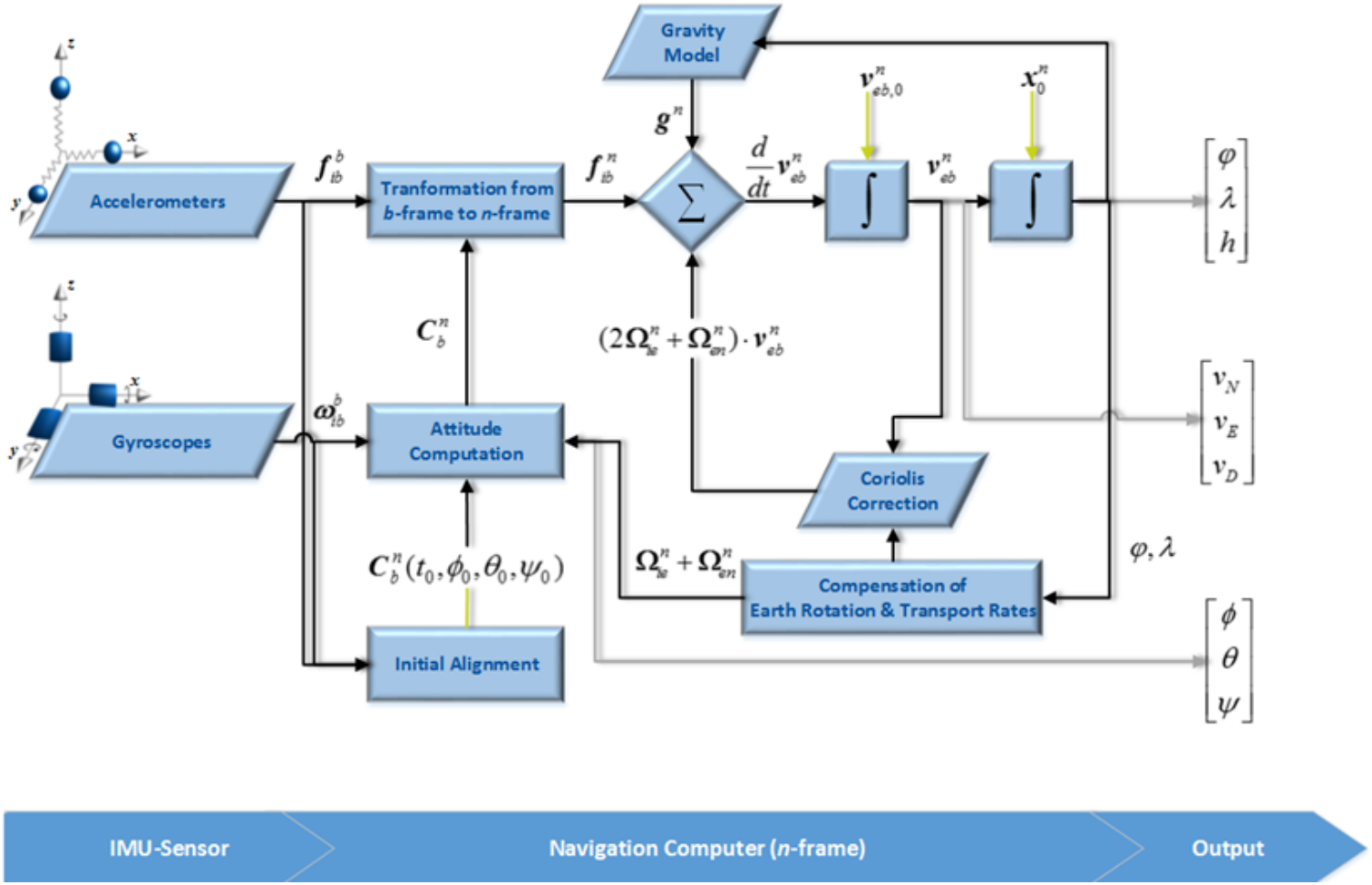
Motivation - Strapdown-Algorithm (Inertial Frame)

Navigation equation: $\ddot{\boldsymbol{x}}^i = \boldsymbol{C}_b^i(\boldsymbol{\omega}_{ib}^i)\boldsymbol{f}_{ib}^b + \boldsymbol{G}^i(\boldsymbol{x}^i)$





Motivation - Strapdown-Algorithm (Navigation Frame)



Tasks 1&2

Task 1:

- a) Define the initial integration constants for velocity and position. Use your static IMU measurement to estimate the Earth rotation rate, g-vector and latitude.
- b) Proceed with an initial alignment (direct DCM calculation) to estimate the initial integration constant for the orientation

Task 2:

- a) Update the attitude, velocity and position of your vehicle frame in the target coordinate system for each IMU epoch from the kinematic dataset. Estimate the final position, velocity and orientation.
  - DCM attitude update, Euler angles for graphic display.
- b) Plot the evolution of the orientation and magnitude of the velocity over time. In a third diagram, show the trajectory of the vehicle on the tangential plane (n-frame).

Experimental Design

Instruments:

- iMAR iIMU-FSAS
  - nav-grade fiber optical gyro
- Automated platform KATE with optical trace sensor

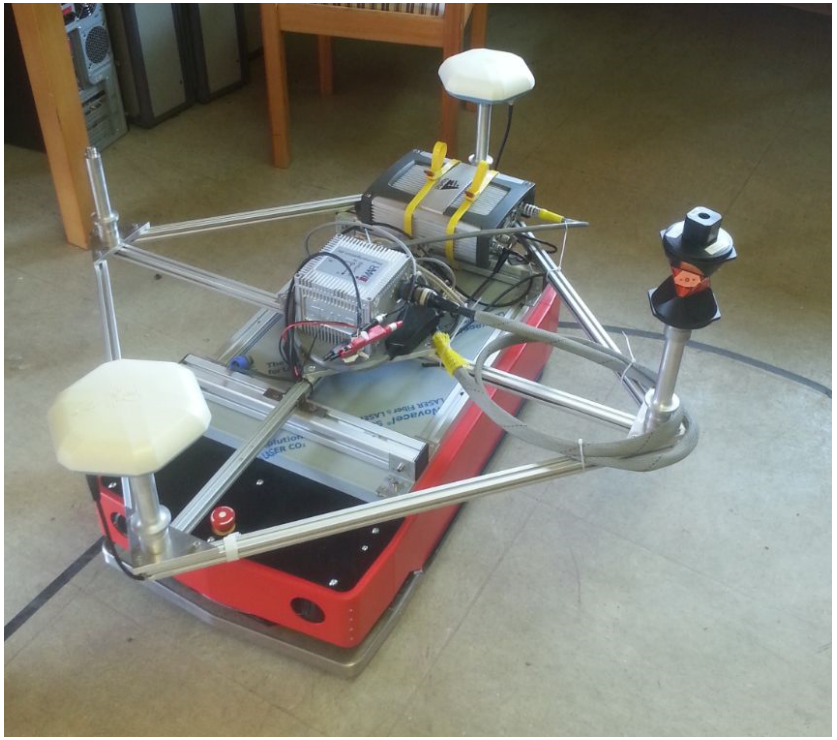
Initialisation time:
$\approx 648\,\mathrm{s}$

Movement time:
$\approx 52\,\mathrm{s}$

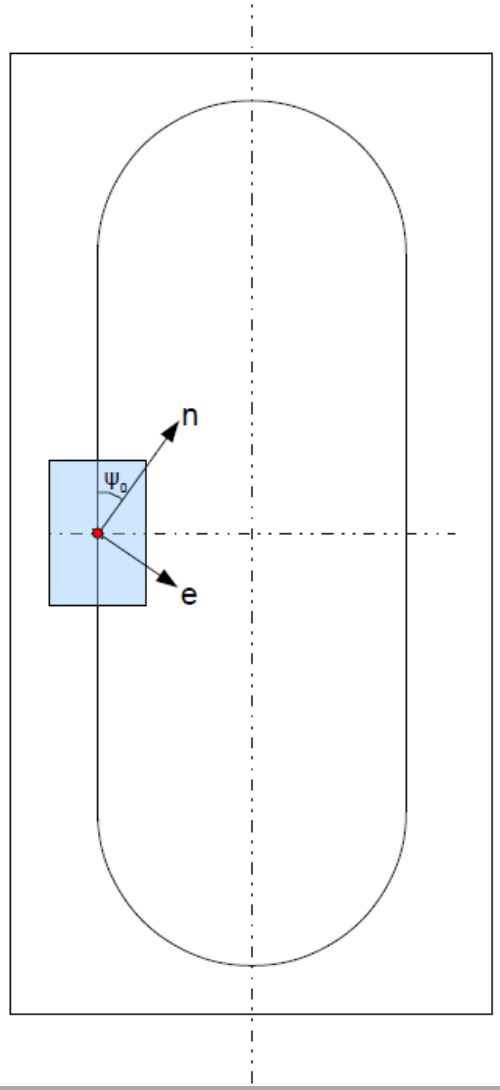
Location: Messdach

Data rate: 200 Hz

Data format: (Matlab *.mat-file)
imudata:
$\{$ time [s], $\boldsymbol{\omega}_{ib}^{b}$ [rad/s], $\boldsymbol{f}_{ib}^{b}$ [m/s^2] $\}$



Reference Trajectory



Elliptic(-ish) track:

- semi-major axis: 4.10 m
- semi-minor axis: 1.60 m
- corner radius: 0.80 m
- Controlled velocity: 1 m/s
- 5 laps

Initial Alignment & Initialisation

Measurement:

- Static during initialisation phase
- IMU is -not- leveled
  - Initial Alignment required
  - use IMU to determine Earth angular rate and latitude first (c.f. lecture slides or prior exercises)
- Steps: $\omega_E \rightarrow \Phi_{lat} \rightarrow C_b^n$
- Re-Orthogonalization?

Assumptions:

- Start in coordinate center of the target system (**n**-frame)
- Velocity during initialisation is zero (vehicle not moving, static alignment)

Navigation algorithm (equations)

Variables:

$\boldsymbol{v}$:	Velocity vector
$\boldsymbol{x}$:	Position vector
$\boldsymbol{g}$:	Gravity vector
$\boldsymbol{\omega}/\boldsymbol{\Omega}$:	Angular rate vector/antisymmetric matrix
$\boldsymbol{f}$:	Specific force vector
$\boldsymbol{C}$:	Rotation matrix/DCM

Differential equations (n-frame, NED-system) (Titterton, 2004):

$$\dot{\boldsymbol{C}}_b^n = \boldsymbol{C}_b^n \cdot \boldsymbol{\Omega}_{nb}^b \tag{3}$$

$$\dot{\boldsymbol{v}}_{eb}^n = \boldsymbol{C}_b^n \boldsymbol{f}_{ib}^b - \left(2\boldsymbol{\omega}_{ie}^n + \boldsymbol{\omega}_{en}^n\right) \times \boldsymbol{v}_{eb}^n + \boldsymbol{g}^n \tag{4}$$

$$\dot{\boldsymbol{x}}_{eb}^n = \boldsymbol{v}_{eb}^n \tag{5}$$

Navigation algorithm (hints)

Attitude update:

- $\boldsymbol{C}_b^n(t+\Delta t) = \boldsymbol{C}_b^n(t)\left[\boldsymbol{I} + \frac{\sin\delta}{\delta}[\boldsymbol{\delta}\times] + \frac{1-\cos\delta}{\delta^2}[\boldsymbol{\delta}\times]^2\right]$ (c.f. ex01b)
- Latitude can be considered constant
- Re-Orthogonalization?

Velocity update:

- Constant value for gravity
- Use trapezoid rule for integration

Position update:

- Integration via trapezoid rule
- Position in cartesian n-Frame (xyz instead of LLH)
  - Initial coordinates can be defined easily.

Performance Analysis

$$\begin{bmatrix} \dot{\boldsymbol{\delta x}} \\ \dot{\boldsymbol{\delta v}} \\ \dot{\boldsymbol{\delta \psi}} \end{bmatrix} = \begin{bmatrix} \mathbf{0}_3 & \boldsymbol{I}_3 & \mathbf{0}_3 \\ \mathbf{0}_3 & -2\boldsymbol{\Omega}_{ie}^n & -[\boldsymbol{f}^n\times] \\ \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{0}_3 \end{bmatrix} \begin{bmatrix} \boldsymbol{\delta x} \\ \boldsymbol{\delta v} \\ \boldsymbol{\delta \psi} \end{bmatrix} + \begin{bmatrix} \mathbf{0}_3 & \mathbf{0}_3 \\ \boldsymbol{I}_3 & \mathbf{0}_3 \\ \mathbf{0}_3 & \boldsymbol{I}_3 \end{bmatrix} \begin{bmatrix} \boldsymbol{\delta f} \\ \boldsymbol{\delta \omega} \end{bmatrix} \tag{6}$$

$\boldsymbol{\delta x}$ - Position error, $\boldsymbol{\delta v}$ - Velocity error, $\boldsymbol{\delta \psi}$ - Attitude error

Task 3:

- a) Compute the theoretical velocity- and position standard deviation based on the white noise densities of the IMU (c.f. iMAR FSAS datasheet).
  Use the given differential equation system and assume constant specific forces (**only gravity**), and constant orientation.
  In this example, the **initial position and velocity** can be regarded as **error free**.
  However, you expect that the **orientation** gained from the initial alignment yields a standard deviation of **1 mrad** on all axes.
  Incorporate this assessment in the initial state covariance matrix. Solve the differential equation system with the *van Loan* method (provided as Matlab function).
  Plot the **magnitude** of the vectors with position and velocity standard deviations over time. Compare the final errors at t=52 s with the results of task 2. Name at least two additional important error sources that were not considered in the theoretical assessment.

Performance Analysis

$$\begin{bmatrix} \dot{\boldsymbol{\delta x}} \\ \dot{\boldsymbol{\delta v}} \\ \dot{\boldsymbol{\delta \psi}} \end{bmatrix} = \underbrace{\begin{bmatrix} \mathbf{0}_3 & \boldsymbol{I}_3 & \mathbf{0}_3 \\ \mathbf{0}_3 & -2\boldsymbol{\Omega}_{ie}^n & -[\boldsymbol{f}^n\times] \\ \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{0}_3 \end{bmatrix}}_{\boldsymbol{A}} \begin{bmatrix} \boldsymbol{\delta x} \\ \boldsymbol{\delta v} \\ \boldsymbol{\delta \psi} \end{bmatrix} + \underbrace{\begin{bmatrix} \mathbf{0}_3 & \mathbf{0}_3 \\ \boldsymbol{I}_3 & \mathbf{0}_3 \\ \mathbf{0}_3 & \boldsymbol{I}_3 \end{bmatrix}}_{\boldsymbol{B}} \begin{bmatrix} \boldsymbol{\delta f} \\ \boldsymbol{\delta \omega} \end{bmatrix} \tag{7}$$

- Reminder: In Ex02, it was possible to express $\boldsymbol{\Phi}_k$, $\boldsymbol{Q}_k$ analytically for any t.
- Generally: no closed analytical solution to strapdown equations possible because specific forces and object rate in n-frame are time-dependent.
- Here: analytical solution possible (thanks to simplifications) but still rather cumbersome
  - skew symmetric terms lead to extensive equations (cf. attitude update).
  - Often, closed solutions are only stated for single channels and under additional conditions (e.g. x and z axis only, see Titterton (2004))
- Numerical approximations often sufficient for short time spans
  - One option: *van Loan* method (Brown and Hwang, 2012).

Performance Analysis :: Van Loan Method

Features:

- Numerical evaluation of $\boldsymbol{\Phi}_k$ and $\boldsymbol{Q}_k$ in one step

$$\boldsymbol{M} = \begin{bmatrix} -\boldsymbol{A}\Delta t & \boldsymbol{B}\boldsymbol{Q}_{\boldsymbol{u}}\boldsymbol{B}^T\Delta t \\ \boldsymbol{0} & \boldsymbol{A}^T\Delta t \end{bmatrix}, \qquad \boldsymbol{N} = \mathrm{expm}(\boldsymbol{M}) = \begin{bmatrix} \dots & \boldsymbol{\Phi}_k^{-1}\boldsymbol{Q}_k \\ \boldsymbol{0} & \boldsymbol{\Phi}_k^T \end{bmatrix} \tag{8}$$

- Matrix exponential solved by Taylor expansion
- Esp. useful for high dimensional systems with low Δt
- See Brown and Hwang (2012), p. 126 for more information.

Computation steps:

- Initialize $\boldsymbol{P}_0$
- $\boldsymbol{\Phi}_k$, $\boldsymbol{Q}_k$ = vanLoan($\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{Q}_{\boldsymbol{u}}, \Delta t$)
  - $\boldsymbol{Q}_{\boldsymbol{u}}$ yields velocity/angular random walk PSD of accelerometers and gyroscopes on main diagonal
  - Just like in exercise 2, but with six entries. See iMAR FSAS data sheet for technical details (iMAR, 2017)
- Propagate $\boldsymbol{P}_{k+1} = \boldsymbol{\Phi}_k\boldsymbol{P}_k\boldsymbol{\Phi}_k^T + \boldsymbol{Q}_k$

Performance Analysis

- b) Which long-term errors (> 1h of free inertial navigation) can be expected from an inertial navigation system in terrestrial applications? Assuming there is a constant initial acceleration bias: do the velocity errors in the horizontal and vertical channel grow unbounded? Give a short statement!
  - Hint: This is independent from the error state model from a). Do errors always grow $\sim t^n$? Are there maybe *sinusoidal* modulations? See Titterton (2004), Chap. 12.

- c) Name and shortly explain an IMU-only technique that can be used to reduce (*re-set*) the velocity drift (other than calibration)
  - Hint: Think about phases in which the vehicle is not moving. Are there techniques that utilize this information?

Analysis

Task 4:

- Plot the transportation rate of the {xyz} channels in a common diagram. Evaluate, if it is really necessary to consider the transportation rate in this experiment.
  Explain which parameters have the biggest impact and state a realistic example in which the transportation rate is expected to be larger than the other systematics (e.g. Earth angular rate).

Transportation rate

- $\boldsymbol{\omega}_{en}^{n} = \begin{pmatrix} v_{eb,E}^{n}/(R_E + h) \\ -v_{eb,N}^{n}/(R_N + h) \\ -v_{eb,E}^{n}\tan(\varphi_{lat})/(R_E + h) \end{pmatrix}$ (Titterton, 2004)
- Simplifications:
  - Earth radius: $R_0 = 6378000[m] \approx R_E \approx R_N$
  - ellipsoidic height: $h = h_0 = 70[m]$

Literature

Literature

- Titterton D, Weston J (2004): "Strapdown Inertial Navigation Technology", 2. Ed., Institution of Electrical Engineers, American Institute of Aeronautics and Astronautics, Peregrinus London.
- Brown, R. G. & Hwang, P. Y. C. (2012). Introduction to random signals and applied Kalman filtering: With MATLAB exercises (4. ed.). Hoboken, NJ: Wiley.
- iMAR. (2017). iIMU-FSAS-NG-EI-SN-E FOG based IMU with trigger and integrated power regulation - special edition for NovAtel SPAN applications only. url: https://www.imar-navigation.de/downloads/IMU_FSAS_NG_EI_SN.pdf
- Groves P.D. (2008): "Principles of GNSS, Inertial, and Multisensor Integrated Navigation Systems", Artech House, Boston.
- Farell J.A. (2008): "Aided Navigation: GPS with High Rate Sensors", McGraw-Hill, New York.